Jayanth Dasamantharao

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WORK EXPERIENCE

AI Engineer - Fedway Associates, *EliteUS Software Solutions* *July 2024 - Present | Edison, NJ*

Developing AI systems that connect multiple agents and deliver real-time insights through **chatbots**, User Interface.

- Built AI agents using AutoGen, LangChain, Azure Search to deliver real-time insights via chat and UI interfaces.
- Developed RAG-based pipelines using OpenAI, LLaMA, Amazon Titan to boost contextual retrieval performance.
- Engineered MLOps pipelines (Kafka, AWS) enabling real-time model re-training, monitoring, and deployment.
- Built full-stack tools like AI chat interfaces (React) and content moderation systems using AWS Rekognition.

Data Science Intern, *Harvest Software Solutions* *June 2023 - Sept 2023 | Jacksonville, Florida*

Worked on advertising project, forecasting ad pricing & analyzing customer sentiment to improve revenue generation.

- Developed forecasting models (XGBoost, LightGBM) and ETL pipelines for ad pricing and sentiment analytics.
- Built NLP pipelines using Hugging Face Transformers, deployed via FastAPI for insights and sentiment analysis.
- Automated model deployment & API workflows, boosting prediction accuracy by 70% reducing manual reporting.

Database Analyst, *Accenture Solutions Pvt. Ltd.* *June 2021 - Aug 2022 | Hyderabad, India*

Client Data Analysis & Data Warehousing, Regeneron Pharmaceuticals Inc.

Managed large healthcare data systems by automating pipelines, maintaining cloud **data warehouse**s, and creating dashboards.

- Built ETL workflows leveraging cloud (Python, AWS Glue) to process 30M+ healthcare records into Redshift & Athena.
- Improved data warehousing performance on AWS (Redshift, Athena, S3), via schema optimization & real-time orchestration.
- Created Tableau/Power BI dashboards to present key metrics, deliver real-time insights for informed stakeholder decisions.

Research Assistant - Statistics, *Rutgers University - NSF* *March 2024 - July 2024 | Jacksonville, Florida*

Built an AI-based self-driving car system using **computer vision and sensor fusion** for lane detection and obstacle recognition.

- Engineered a Puppeteer-based web scraping system, capturing information of 500K+ lawyer profiles with 99% accuracy.
- Built statistical pipelines using (Z-score, IQR) to detect anomalies and validate legal data trends.
- Applied regression, clustering, and time-series forecasting to uncover profession-level insights.

Researcher - Image Processing & Deep Learning, *Andhra University* *Jan 2021 - May 2021 | Visakhapatnam, India*

Built an AI-based self-driving car system using **computer vision and sensor fusion** for lane detection and obstacle recognition.

- Developed an autonomous driving system using YOLO, Faster R-CNN, CNNs for object detection and lane tracking.
- Integrated multi-sensor data (LiDAR, GPS, radar) for obstacle recognition and environment awareness. [Research Paper.]
- Built Unity3D simulation with real-time Flask backend for parameter monitoring and model validation.

TECHNICAL SKILLS

Programming: Python, R, Scala, SQL. **Libraries:** Scikit-learn, PyTorch, TensorFlow, FastAPI, LangChain, AutoGen.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Database & Cloud: Docker, Kafka, AWS(S3, Glue, RDS, Lambda), GCP, ETL, BigQuery, MongoDB, Langchain, MLOps.
Focus Areas: RAG Pipelines, LLM's, Recommendation Engines, Distributed Systems, Backend APIs, Real-time AI.

EDUCATION

Rutgers University, *Masters in Data Science* *Sept 2022 - May 2024 | New Brunswick, NJ*

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** Portfolio.

Andhra University, *B.Tech in Electrical and Electronics Engineering* *June 2017 - June 2021 | Visakhapatnam, India*

Relevant Coursework: Python, Data Structures & Algorithms, Database Management, Operating Systems, System Design.

PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Twitter Search Application - [Database Management, Python, Sorting techniques, Search Engine Optimization] Created a user query-responsive tweet retrieval system by integrating Postgres for user data management and MongoDB for tweet storage. Optimized data storage and retrieval efficiency by 35% while implementing caching strategies for system performance.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.